

Kiduka Soil Status Prediction and Fertilizer Recommendation:

1. Executive Summary

Kiduka is a sophisticated, data-driven agricultural decision-support system designed to optimize farming outcomes by providing precise soil health insights and localized resource recommendations. By bridging the gap between advanced environmental satellite data and actionable agricultural practices, Kiduka empowers farmers to make scientifically backed decisions. The system's defining feature is its highly adaptable triple-mode architecture, which seamlessly accommodates users regardless of their access to formal laboratory soil testing, providing accurate fertility predictions based on varying levels of available data.

2. Triple Prediction Mode Architecture

To ensure maximum accessibility, precision, and flexibility, Kiduka has transitioned from a dual-engine to a dynamic triple-mode framework. The system intelligently routes and merges data through Machine Learning (ML), Hybrid, or Traditional Formula-Based modes depending on the user's inputs.

A. Machine Learning (ML) Predictive Engine

Designed for rapid, accessible analysis when formal lab results are unavailable.

When a user inputs only their GPS coordinates and current soil pH level, Kiduka activates its ML predictive engine. This engine integrates directly with the Google Earth Engine (GEE) to fetch high-resolution, satellite-derived environmental features, including:

- **Vegetation Indices (NDVI):** Sourced from MODIS imagery to evaluate plant health trends.
- **Soil Texture:** Extracted from OpenLandMap to determine the sand-to-clay ratio.
- **SAR (Radar) Data:** Sentinel-1 Synthetic Aperture Radar (VV and VH bands) utilized to assess surface roughness and soil moisture.

- **Terrain & Climate Analysis:** Elevation and slope data from SRTM, combined with historical rainfall patterns (total, dry season, and short/long rains) from the CHIRPS dataset.

These variables are processed through advanced Neural network models to estimate critical soil macronutrients and micronutrients: Nitrogen (N), Phosphorus (P), Potassium (K), Organic Carbon (OC), Calcium (Ca), and Magnesium (Mg). The system then computes a weighted Soil Health Index (SHI), classifying the soil fertility into actionable tiers: Healthy, Moderately Healthy, Poor, or Very Poor.

B. Hybrid Prediction (ML-Enhanced Analysis)

Designed for maximum data utility when partial laboratory results are available.

This mode is triggered when a user provides their GPS coordinates, pH, and some elective nutrients (e.g., they know their Nitrogen and Phosphorus levels but lack the rest).

- **Gap Filling:** The system calls the ML engine to estimate only the nutrients that were left blank by the user.
- **Data Merging & Override:** User-provided measurements always override ML predictions. The AI's estimations for provided values are strictly discarded in favor of actual lab data.
- **SHI Recalculation:** The system explicitly discards the ML-predicted overarching SHI. Instead, it computes a new, highly accurate SHI using the unified set of measured data and AI-estimated gap-fills. This ensures the final status is firmly anchored by the user's actual lab data.

C. Formula-Based Diagnostic Engine

Designed for high-precision, rule-based analysis utilizing laboratory data.

When a user inputs specific nutrient values from a laboratory soil test, the system shifts to a traditional diagnostic model. This model utilizes:

1. **Scientific Weighting:** Parameters are assigned scientific weights (e.g., pH: 3.0, OC: 2.5) to generate a composite Soil Health Index.

2. **Expert System Override Rules:** To prevent misleading composite scores, the classifier applies strict scientific logic:
 - Rule 1 (pH Cap): A "Very Poor" pH level restricts the maximum overall status to "Poor."
 - Rule 2 (OC Downgrade): A "Very Poor" Organic Carbon level automatically downgrades the overall status by one class.
 - Rule 3 (Threshold Rule): If three or more individual parameters are flagged as "Very Poor," the overarching status is forced to "Very Poor."
 - Rule 4 (Macro Rule): If pH is "Poor" and two or more Macronutrients (N, P, K) are "Poor" or "Very Poor," the overall status is capped at "Poor."

3. Agrovet Recommendation Engine

Kiduka extends beyond passive analysis by facilitating direct access to required agricultural inputs. The built-in recommendation engine uses geospatial intelligence to connect farmers with local suppliers.

1. **Geospatial Targeting:** Utilizing the Haversine formula, the system accurately calculates spherical distances to identify the closest Agrovets within a 100km radius.
2. **Nutrient-Based Product Matching:** Kiduka dynamically aligns the user's predicted or measured nutrient deficiencies with specific fertilizer recommendations. For example:
 - a. **Nitrogen Deficit:** Recommends Calcium Ammonium Nitrate (CAN) for top-dressing.
 - b. **Phosphorus Deficit:** Recommends NPK or DAP fertilizers at the planting stage.
 - c. **Organic Carbon Deficit:** Recommends applying Farm Yard Manure (FYM) or compost.
3. **Proximity Ranking:** The system outputs a curated, ranked list of the five nearest Agrovets equipped to supply the specifically recommended inputs, streamlining the procurement process for the farmer.

4. User Interface & Experience Design

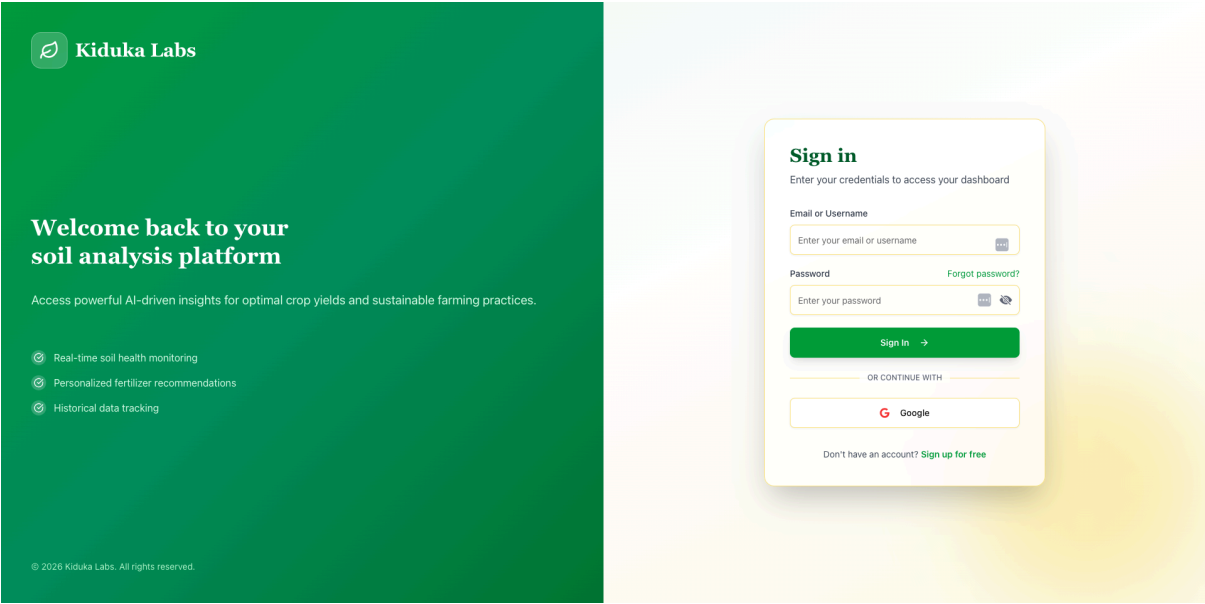
Kiduka is engineered with a user-centric interface that prioritizes simplicity, ensuring farmers can easily navigate the workflow from input to action.

1. **Dynamic Input Workflow:** The entry form is segmented logically. The Basic section (GPS and pH) serves as the minimum requirement to trigger the ML engine. The Nutrients section is optional; populating it partially triggers the Hybrid engine, while populating it fully triggers the high-precision Formula-based engine.
2. **Comprehensive Results Dashboard:** Upon analysis, users are presented with an intuitive dashboard featuring:
 1. A clear, visual indicator of overall Soil Fertility Status.
 2. The precise numerical Soil Health Index (SHI) Score.
 3. Mode Awareness text indicating if the results are from "Machine Learning," "Hybrid Analysis," or "Standard Formula."
 4. Granular visual scores for individual nutrient breakdowns, complete with "Measured" or "Estimated" data provenance badges.
 5. Actionable, personalized fertilizer application instructions.
 6. An interactive, map-integrated list of the nearest relevant Agrovets.

User Interface Flow:

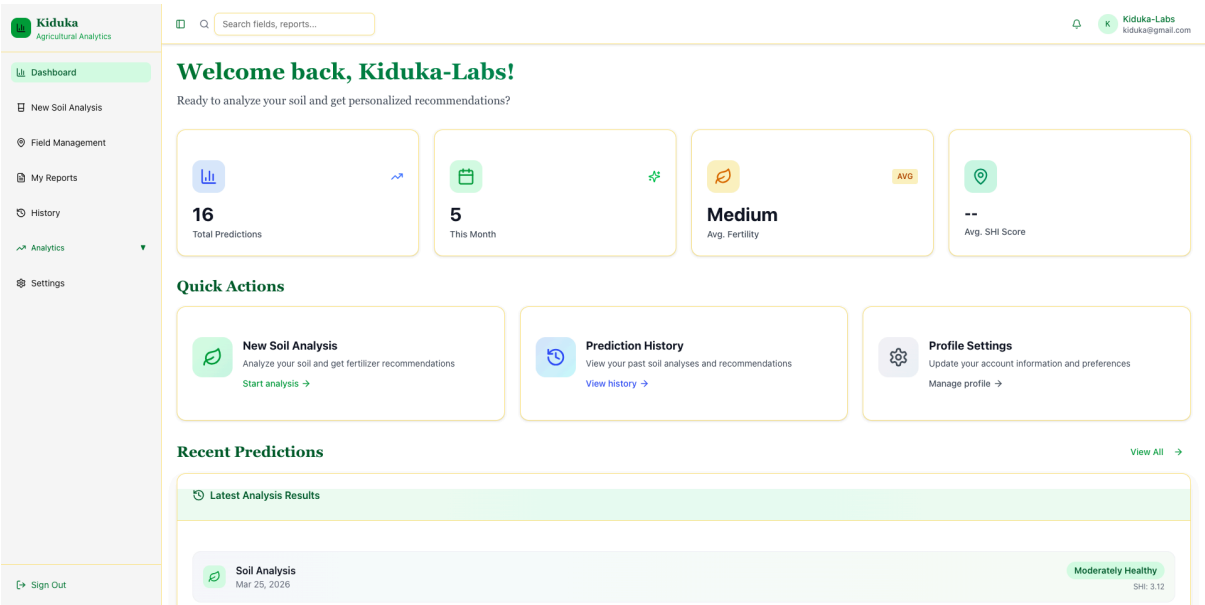
1. Signup/Login page:

A secure entry point featuring mobile number or email authentication. It includes straightforward options for "Create Account" and "Forgot Password" to ensure accessibility for all farming demographics.



2. Main Dashboard page:

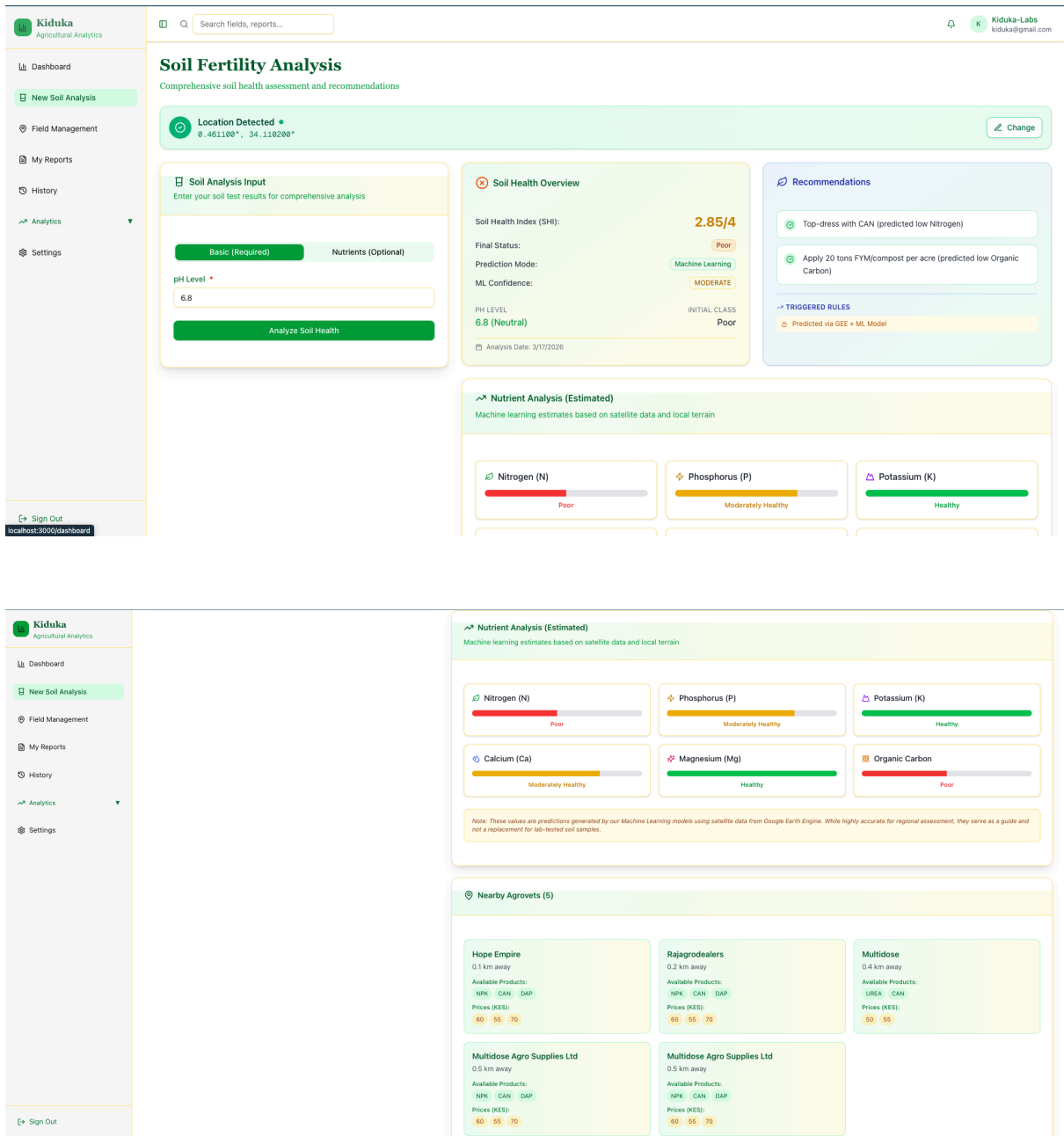
The central hub for the user. It features a welcoming overview, a summary of the most recent soil health status and a prominent "Start New Analysis" call-to-action button.



3. Soil Analysis Page:

A. Machine Learning Prediction Mode:

A clean input form requiring only two fields: an auto-detectable "GPS Location" pin and a manual input field for "Soil pH." Clicking "Analyze Soil" fetches the satellite data and runs the Neural Network models.



B. Hybrid Based Prediction:

An adaptable form where users input GPS, pH, and some lab metrics. The system processes the partial data, estimating missing fields in the background, and seamlessly displays the merged results on the dashboard using provenance badges:

Kiduka

Agricultural Analytics

Dashboard

New Soil Analysis

Field Management

My Reports

History

Analytics

Settings

Sign Out

Search fields, reports...

Soil Fertility Analysis

Comprehensive soil health assessment and recommendations

Location Detected

0.451100°, 34.110200°

Change

Soil Analysis Input

Enter your soil test results for comprehensive analysis

Basic (Required)

Nutrients (Optional)

Organic Carbon (%)

(Optional)

2.7

Nitrogen (N) (%)

(Optional)

0.20

Phosphorus (P) (ppm)

(Optional)

35.2

Potassium (K) (ppm)

(Optional)

180.0

Calcium (Ca) (ppm)

(Optional)

1250

Magnesium (Mg) (ppm)

(Optional)

220

Soil Health Overview

Soil Health Index (SHI): 3.27/4

Final Status: Moderately Healthy

Prediction Mode: Hybrid Analysis

User Input + AI Prediction

ML Confidence: MODERATE

PH LEVEL: 6.5 (Neutral)

INITIAL CLASS: Moderately Healthy

Analysis Date: 9/25/2026

Recommendations

Top-dress with CAN

TRIGGERED RULES

Hybrid: User Input + ML Prediction

Hybrid Soil Analysis

Results combine your measured inputs with ML-predicted estimates for missing values.

Nitrogen (N)

Estimated

Poor

Phosphorus (P)

Estimated

Moderately Healthy

Potassium (K)

Estimated

Healthy

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(Optional)

220

Analyze Soil Health

Soil Health Overview

Soil Health Index (SHI): 3.27/4

Final Status: Moderately Healthy

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User Input + AI Prediction

ML Confidence: MODERATE

PH LEVEL: 6.5 (Neutral)

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TRIGGERED RULES

Hybrid: User Input + ML Prediction

Hybrid Soil Analysis

Results combine your measured inputs with ML-predicted estimates for missing values.

Nitrogen (N)

Estimated

Poor

Phosphorus (P)

Estimated

Moderately Healthy

Potassium (K)

Estimated

Healthy

Calcium (Ca)

Measured

1245.0 ppm

Magnesium (Mg)

Estimated

Healthy

Organic Carbon

Measured

2.7 %

Optimal Range Guide

Below Optimal - Deficiency

Optimal Range - Ideal for plant growth

Above Optimal - High reserves

Note: Estimated values are predictions generated by our Machine Learning models using satellite data from Google Earth Engine. Your measured inputs are prioritized.

Nearby Agrovets (5)

Hope Empire

0.1 km away

Available Products: NPK, CAN, DAP

Prices (KES): 60, 55, 70

Rajagrodealers

0.2 km away

Available Products: NPK, CAN, DAP

Prices (KES): 60, 55, 70

Multidose

0.4 km away

Available Products: UREA, CAN

Prices (KES): 50, 55

Multidose Agro Supplies Ltd

0.5 km away

Available Products: NPK, CAN, DAP

Prices (KES): 60, 55, 70

Multidose Agro Supplies Ltd

0.5 km away

Available Products: NPK, CAN, DAP

Prices (KES): 60, 55, 70

C. Formula Based Prediction:

An expanded form that reveals input fields for specific lab metrics (N, P, K, OC, Ca, Mg). Users input all exact laboratory numbers here, completely bypassing the ML estimations in favor of the rule-based expert system.

 Kiduka

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Search fields, reports...

Soil Fertility Analysis

Comprehensive soil health assessment and recommendations

Location Detected

9.461108°, 34.118288°

Change

Soil Analysis Input

Enter your soil test results for comprehensive analysis

Basic (Required)

Nutrients (Optional)

Organic Carbon (%) (Optional)

2.5

Nitrogen (N) (%) (Optional)

0.2

Phosphorus (P) (ppm) (Optional)

38

Potassium (K) (ppm) (Optional)

190

Calcium (Ca) (ppm) (Optional)

1250

Magnesium (Mg) (ppm) (Optional)

220

Analyze Soil Health

Soil Health Overview

Soil Health Index (SHI): 3.35/4

Final Status: Moderately Healthy

Prediction Mode: Standard Formula

PH LEVEL: 6.8 (Neutral)

INITIAL CLASS: Moderately Healthy

Analysis Date: 3/17/2028

Recommendations

Maintain current practices

Nutrient Analysis

Visual analysis of soil nutrient levels with optimal ranges

Nitrogen (N)

0.2 %

0.15 %

Optimal Range

0.3 %

Phosphorus (P)

38.0 ppm

20 ppm

Optimal Range

50 ppm

Potassium (K)

190.0 ppm

80 ppm

Optimal Range

200 ppm

Calcium (Ca)

1250.0 ppm

1000 ppm

Optimal Range

2500 ppm

Magnesium (Mg)

220.0 ppm

150 ppm

Optimal Range

350 ppm

Organic Carbon

2.5 %

2 %

Optimal Range

4 %

 Kiduka

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38

Potassium (K) (ppm) (Optional)

190

Calcium (Ca) (ppm) (Optional)

1250

Magnesium (Mg) (ppm) (Optional)

220

Analyze Soil Health

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Visual analysis of soil nutrient levels with optimal ranges

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0.2 %

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Potassium (K)

190.0 ppm

80 ppm

Optimal Range

200 ppm

Calcium (Ca)

1250.0 ppm

1000 ppm

Optimal Range

2500 ppm

Magnesium (Mg)

220.0 ppm

150 ppm

Optimal Range

350 ppm

Organic Carbon

2.5 %

2 %

Optimal Range

4 %

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Available Products: NPK CAN DAP

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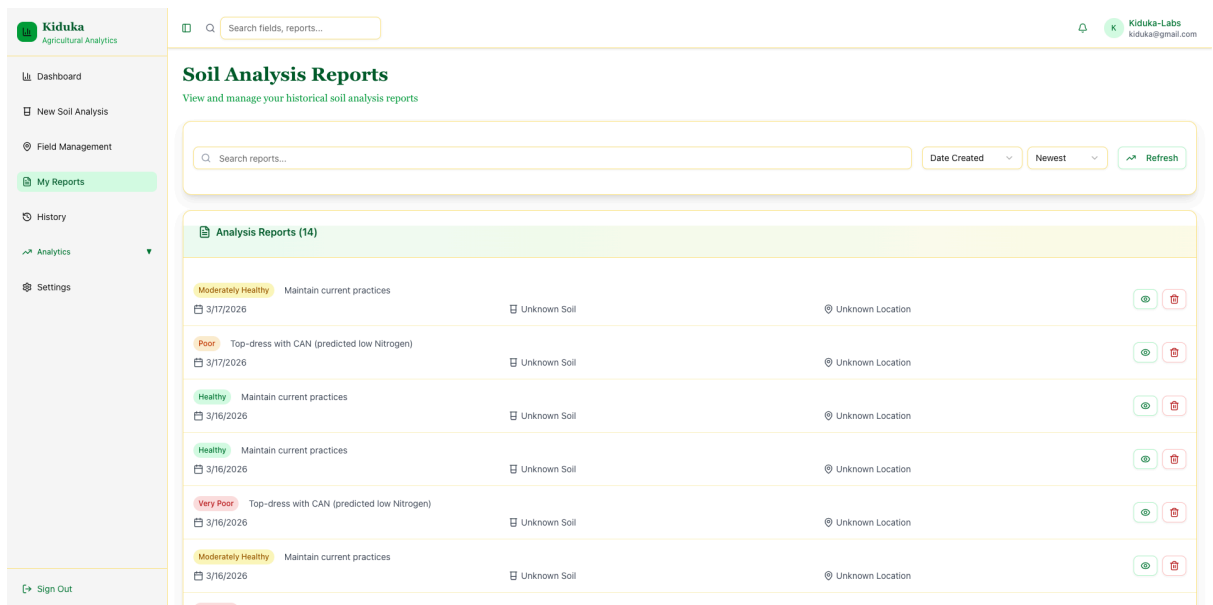
0.5 km away

Available Products: NPK CAN DAP

Prices (KES): 60 55 70

4. Reports History Page

A chronological, easily searchable log of all past soil tests.



5. Conclusion

Kiduka represents a highly scalable and scientifically rigorous solution to soil fertility management. By intelligently harmonizing localized lab data with global satellite imagery through its triple-mode architecture, and pairing diagnostics with a location-aware supply chain recommendation system, Kiduka provides an end-to-end operational tool. It is uniquely positioned to assist modern farmers in improving crop yields, optimizing fertilizer expenditure, and maintaining long-term soil health.